

NON-PARAMETRIC ESTIMATION OF SURVIVAL CURVES

Example: with the following survival data, calculate the survival function $S(t)$ by using the Kaplan & Meier and Nelson-Aalen estimators. The follow up finish at $t = 35$.

Time	Exposition	# events
6	21	3
7	17	1
10	15	1
13	12	1
16	11	1
22	7	1
23	6	1

Kaplan and Meier estimator:

t_i	d_i	R_i	$\hat{S}(t)$	$\sum_{t_j \leq t} \frac{d_j}{R_i(R_i - d_j)}$	$\hat{S}(t)^2 \sum_{t_j \leq t} \frac{d_j}{R_i(R_i - d_j)}$
6	3	21	$[1 - \frac{3}{21}] = 0.857$	$\frac{3}{21 \times 18} = 0.0079$	$0.857^2 \times 0.0079 = 0.0058$
7	1	17	$[0.857](1 - \frac{1}{17}) = 0.807$	$0.0079 + \frac{1}{17 \times 16} = 0.0116$	$0.807^2 \times 0.0116 = 0.0076$
10	1	15	$[0.807](1 - \frac{1}{15}) = 0.753$	$0.0116 + \frac{1}{15 \times 14} = 0.0164$	$0.753^2 \times 0.0164 = 0.0093$
13	1	12	$[0.753](1 - \frac{1}{12}) = 0.690$	$0.0164 + \frac{1}{12 \times 11} = 0.0240$	$0.690^2 \times 0.0240 = 0.0114$
16	1	11	$[0.690](1 - \frac{1}{11}) = 0.628$	$0.0240 + \frac{1}{11 \times 10} = 0.0330$	$0.628^2 \times 0.0330 = 0.0130$
22	1	7	$[0.628](1 - \frac{1}{7}) = 0.538$	$0.0330 + \frac{1}{7 \times 6} = 0.0569$	$0.538^2 \times 0.0569 = 0.0164$
23	1	6	$[0.538](1 - \frac{1}{6}) = 0.448$	$0.0569 + \frac{1}{6 \times 5} = 0.0902$	$0.448^2 \times 0.0902 = 0.0181$

Time on Study (t)	$\hat{S}(t)$	Standard Error
$0 \leq t < 6$	1.000	0.000
$6 \leq t < 7$	0.857	0.076
$7 \leq t < 10$	0.807	0.087
$10 \leq t < 13$	0.753	0.096
$13 \leq t < 16$	0.690	0.107
$16 \leq t < 22$	0.628	0.114
$22 \leq t < 23$	0.538	0.128
$23 \leq t < 35$	0.448	0.135

Nelson-Aalen estimator:

t	$\hat{\Lambda}(t) = \sum_{t_j \leq t} \frac{d_j}{R_i}$	$\sigma^2 = \sum_{t_j \leq t} \frac{d_j}{R_i^2}$	Standard error
$0 \leq t < 6$	0	0	0
$6 \leq t < 7$	$\frac{3}{21} = 0.1428$	$\frac{3}{21^2} = 0.0068$	0.0825
$7 \leq t < 10$	$0.1428 + \frac{1}{17} = 0.2017$	$0.0068 + \frac{1}{17^2} = 0.0103$	0.1015
$10 \leq t < 13$	$0.2017 + \frac{1}{15} = 0.2683$	$0.0103 + \frac{1}{15^2} = 0.0147$	0.1212
$13 \leq t < 16$	$0.2683 + \frac{1}{12} = 0.3517$	$0.0147 + \frac{1}{12^2} = 0.0217$	0.1473
$16 \leq t < 22$	$0.3517 + \frac{1}{11} = 0.4426$	$0.0217 + \frac{1}{11^2} = 0.0299$	0.1729
$22 \leq t < 23$	$0.4426 + \frac{1}{7} = 0.5854$	$0.0299 + \frac{1}{7^2} = 0.0503$	0.2243
$23 \leq t < 35$	$0.5854 + \frac{1}{6} = 0.7521$	$0.0503 + \frac{1}{6^2} = 0.0781$	0.2795

t	$\exp[-\hat{\Lambda}(t)]$
$0 \leq t < 6$	1.0000
$6 \leq t < 7$	0.8669
$7 \leq t < 10$	0.8173
$10 \leq t < 13$	0.7647
$13 \leq t < 16$	0.7035
$16 \leq t < 22$	0.6424
$22 \leq t < 23$	0.5569
$23 \leq t < 35$	0.4714

